Trigonometry & Additional Topics

ANSWER KEY

Right-triangle trigonometry

1 In the right triangle shown below, the hypotenuse is 20 and the side opposite angle θ is 12. Find the length of the adjacent side, then find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Adjacent $=\sqrt{20^2-12^2}=\sqrt{400-144}=\sqrt{256}=16$. $\sin\theta=\frac{12}{20}=\frac{3}{5}$. $\cos\theta=\frac{16}{20}=\frac{4}{5}$. $\tan\theta=\frac{12}{16}=\frac{3}{4}$.

2 In a right triangle, angle θ has adjacent side 7 and hypotenuse 25. Find $\sin\theta$ and $\tan\theta$.

Opposite $=\sqrt{25^2-7^2}=\sqrt{625-49}=\sqrt{576}=24$. $\sin\theta=\frac{24}{25}$. $\tan\theta=\frac{24}{7}$.

3 A 15-foot ladder leans against a wall, making a 50° angle with the ground. Find how high up the wall the ladder reaches, to the nearest tenth of a foot.

Height $=15\sin(50^\circ)\approx 15(0.766)\approx 11.5$ feet.

The Pythagorean identity

4 Given $\sin\theta=\frac{5}{13}$ with θ acute, find $\cos\theta$ using $\sin^2\theta+\cos^2\theta=1$.

$\cos^2\theta=1-\frac{25}{169}=\frac{144}{169}$. Since θ is acute, $\cos\theta=\frac{12}{13}$.

5 Given $\cos\theta=\frac{8}{17}$ with θ acute, find $\sin\theta$.

$\sin^2\theta=1-\frac{64}{289}=\frac{225}{289}$. Since θ is acute, $\sin\theta=\frac{15}{17}$.

Exact trigonometric values

6 State the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$.

$\sin\left(\frac{\pi}{6}\right)=\frac{1}{2}$. $\cos\left(\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$.

7 State the exact value of $\tan(60^\circ)$ and $\sin(90^\circ)$.

$\tan(60^\circ)=\sqrt{3}$. $\sin(90^\circ)=1$.

Solving for missing sides and angles

8 A surveyor stands 150 feet from the base of a tower and measures the angle of elevation to the top as 32°, as shown below. Find the height of the tower, to the nearest foot.

$h=150\tan(32^\circ)\approx 150(0.6249)\approx 94$ feet.

9 From the top of a 60-foot cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

The angle of depression equals the angle of elevation from the boat, 25°, so $\tan(25^\circ) = \frac{60}{d}$, giving $d = \frac{60}{\tan(25^\circ)} \approx \frac{60}{0.4663} \approx 129$ feet.

Complex numbers: arithmetic

10 Add $(4+3i)+(2-7i)$.

$6-4i$.

11 Multiply $(3+2i)(1-4i)$.

$3-12i+2i-8i^2 = 3-10i-8(-1) = 11-10i$.

12 Simplify i^7, using $i^2 = -1$.

$i^7 = i^4 \cdot i^3 = 1 \cdot i^3 = i^2 \cdot i = -1 \cdot i = -i$.

Complex numbers: modulus and conjugates

13 Find the modulus $|6-8i|$.

$\sqrt{6^2+8^2} = \sqrt{36+64} = \sqrt{100} = 10$.

14 Find the complex conjugate of $-3+5i$, and compute the product of the number and its conjugate.

Conjugate: $-3-5i$. Product $= (-3+5i)(-3-5i) = 9-25i^2 = 9+25 = 34$.

Vectors: magnitude and operations

15 Vector $v = (3,4)$. Find its magnitude.

$|v| = \sqrt{3^2+4^2} = \sqrt{9+16} = \sqrt{25} = 5$.

16 Given vectors $u = (2,-5)$ and $v = (-6,3)$, find $u+v$.

$u+v = (2+(-6), -5+3) = (-4,-2)$.

17 Given vector $w = (8,-6)$, find $2w$ and the magnitude of $2w$.

$2w = (16,-12)$. $|2w| = \sqrt{16^2+12^2} = \sqrt{256+144} = \sqrt{400} = 20$.

Radian measure

18 Convert 150° to radians, in terms of π.

$150 \times \frac{\pi}{180} = \frac{5\pi}{6}$.

19 Convert $\frac{7\pi}{4}$ radians to degrees.
$\frac{7\pi}{4} \times \frac{180}{\pi} = 7 \times 45^\circ = 315^\circ$.

A well-designed word problem: measuring the width of a river

20 This question has three parts. A surveyor stands at point A directly across a river from a tree at point B (so angle $A = 90^\circ$), then walks 80 meters along the riverbank to point C. (a) At point C, the angle between the bank and the line of sight to the tree is 35°. Use the tangent ratio to find AB, the width of the river, to the nearest meter. (b) Find the direct distance CB from C to the tree, to the nearest meter. (c) A colleague measures from a point 50 meters past C (so 130 meters from A) and finds the angle to the tree is 24°. Use this second measurement to find a second estimate of AB, to the nearest meter, and state which measurement gives the larger estimate.
(a) $\tan(35^\circ) = \frac{AB}{80}$, so $AB = 80\tan(35^\circ) \approx 80(0.7002) \approx 56$ meters. (b) $CB = \frac{80}{\cos(35^\circ)} \approx \frac{80}{0.8192} \approx 98$ meters. (c) $AB = 130\tan(24^\circ) \approx 130(0.4452) \approx 58$ meters. The second measurement (58 m) is slightly larger than the first (56 m); the two estimates are reasonably consistent.